Tri Huu Nguyen

Backend Software Engineer
Ho Chi Minh City · +84 94 342 1511 · nguyenhuutri1620vn@gmail.com
LinkedIn: linkedin.com/in/tri-nguyen-huu · Git: github.com/nguyenhuutri1620vn

SUMMARY

Backend engineer with 4+ years of experience building data-heavy SaaS products for healthcare supply chain. Strongest in Go: multi-tenant PostgreSQL for 19+ organizations, sub-second Elasticsearch analytics, Kafka-based large exports, and ClickHouse pipelines behind Superset dashboards. Also ships production Ruby on Rails features in a 12-person product team. Comfortable with complex business rules such as contracts, rebates, and price comparison. Use Cursor and Claude Code to self-review code and catch issues early.

TECHNICAL SKILLS

Backend: Golang, REST APIs, Echo v4, Ruby, Ruby on Rails, Clean Architecture, Kafka.
Databases: PostgreSQL, ClickHouse, Elasticsearch.
AI tools: Cursor, Claude Code.
Reporting/BI: Apache Superset.
Infrastructure: Docker, AWS (S3, Redshift, Secrets Manager).
Frontend: React, Ant Design, Typescript.

PROFESSIONAL EXPERIENCE

Company: BESTARION | **Position**: Sofware Engineer (Backend focus) Feb 2022 - Present

Team SSC | Insight Dashboard, Value Analysis, Custom Report Builder

Backend for a multi-tenant healthcare spend analytics SaaS platform that helps hospitals track procurement spending, vendor performance, and cost savings.

Technologies: Golang, PostgreSQL, Elasticsearch, ClickHouse, Kafka, AWS, Superset, Docker.

- Designed a multi-tenant architecture with a separate PostgreSQL database per organization, isolating data for 19+ active healthcare organizations.
- Built the backend of the Value Analysis product: converts units and pack sizes to a common each price so items from different vendors and manufacturers can be compared, and calculates potential savings and profit/loss for a group of items.
- Developed and maintained 90+ REST APIs for spend, vendor, contract compliance, price variance, and value analysis features.
- Engineered a search and aggregation layer with Elasticsearch, returning spend analytics filtered by period, corporation, and vendor in under 1 second.
- Implemented Kafka-based async workers for Excel exports with streaming writes and automatic retries; exports of 1M+ rows are split into multiple files and zipped, eliminating timeouts.
- Built ETL pipelines from PostgreSQL through S3 into ClickHouse (full and incremental loads), creating the data models behind Superset dashboards for spend and trend analysis.
- Automated Superset dashboard rollout across 19+ organizations by cloning dashboards, charts, and datasets instead of rebuilding them by hand.
- Built core backend services in Golang following Clean Architecture, separating business logic from infrastructure through repository interface abstractions.

- Structured services with Clean Architecture; applied goroutines, worker pools, bulk loading, transactions, and query tuning for high-volume data processing.
- Used AWS S3 for data archiving, Secrets Manager for credentials, and Redshift for warehouse integration.
- Designed and maintained REST APIs consumed by React front-end applications, ensuring backward compatibility across releases.

Team DEV | Healthcare Supply Chain System Platform | **Position**: Fullstack Developer
Built and maintained features for Meperia MSSS, a multi-tenant SaaS platform that helps hospitals standardize medical supply data, manage contracts, and track spending.
Technologies: Ruby, Ruby on Rails, PostgreSQL, Solr, AWS, ExtJS.

- Engineered the Future Price Discrepancy module, enabling hospital administrators to cross-compare upcoming vs. active contract prices across facilities and bulk-resolve discrepancies prior to activation, proactively preventing spend leakages.
- Developed the core Contract Rebate Management engine, automating volume-based and tiered rebate tracking against billing-cycle spend baselines to surface real-time eligibility and maximize financial recovery.
- Built the contract archival pipeline: moves contracts expired for 3+ years to AWS S3 in safe batches with row-count checks, keeping the main database fast and Superset reports complete.
- Strengthened business logic reliability through comprehensive automated testing, covering complex multi-facility approval workflows, tiered rebate calculations, and lifecycle state transitions to eliminate regression risks.
- Designed an AI-assisted pull request quality gate (static analysis, security scan, dependency scan, AI plus human review) with checklists for code and unit tests, reducing review effort.

EDUCATION

Thu Dau Mot University | Bachelor's degree in Software Engineering | GPA: 3.0/4.0
Duration: Oct 2018 - Jun 2022

CERTIFICATION

AWS Certified Cloud Practitioner (CLF-C02) | Score: 811 | Link: Credly